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Q. 3: A radionuclide has a half life of 10 s. Which of the following statements are correct?

(a) In a sample, probability that a nucleus will decay in next 10 s is 0.5. ✓

(b) In a sample, the probability that a nucleus which has survived first 10 s, will decay in next 10 s is 0.5. ✓

(c) The probability that a nucleus which has survived first 10 s, will decay in next 10 s is 0.25.

(d) If a sample of radionuclide has 4 nuclei, two nuclei will decay in next 10 s. ×

$T_{\frac{1}{2}} = 10\,sec$

$\frac{(N_0/2)}{(N_0)} = \frac{1}{2} = 0.5$

$\frac{N_0/4}{N_0/2} = 0.5$

	A $\longrightarrow$ B
$t=0$	N_0 4 0
$t=T_{\frac{1}{2}}$	$\frac{N_0}{2}$ 2
$t=2T_{\frac{1}{2}}$	$(N_0/4)$

$\frac{N_0}{2}$

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Q. 12: Consider the nuclear reaction

$$^7_3Li + ^1_1H \longrightarrow 2\ ^4_2He$$

The relevant atomic masses are:

$^7Li: 7.018$ u; $^1H: 1.008$ u; $^4He: 4.004$ u.

One gram of ^1H is completely consumed in the reaction along with sufficient amount of ^7Li. Find energy released.

$$_3Li^7 + {_1H^1} \longrightarrow 2\,_2He^4 + Q$$

$$\Delta m = M_{Li} + M_H - 2M_{He}$$

$$= 7.018 + 1.008 - 8.008$$

$$= 0.018 \text{ amu}$$

S.I. $Q = 0.018 \times 1.67 \times 10^{-27} \times (3 \times 10^8)^2$ Joules

1 gm

$1 \text{ gm} \longrightarrow N_A = 6.023 \times 10^{23}$

$$Q_{net} = 0.018 \times 1.67 \times 10^{-27} \times 9 \times 10^{16} \times 6.023 \times 10^{23} \text{ Joule}$$

$$= 1.62 \times 10^{12} \text{ Joule}$$

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Q. 21: Nuclei of ^{64}Cu can decay by electron capture (probability 61%) or by β^+ decay (39%). Half life of ^{64}Cu is 12.7 hour. Find the partial half life for electron capture decay process.

$$Cu \xrightarrow{\lambda_{EC}} X$$
$$Cu \xrightarrow{\lambda_\beta} Y$$

N

$$\frac{dN}{dt} = -\lambda_{EC}N - \lambda_\beta N$$
$$= -\underbrace{(\lambda_{EC} + \lambda_\beta)}N$$

$$\lambda = \lambda_{EC} + \lambda_\beta$$

$$\lambda_\beta = \frac{39}{61}\lambda_{EC}$$

$$\frac{\lambda_{EC}\cancel{N}}{\lambda_\beta \cancel{N}} = \frac{61}{39}$$

$$\frac{\ln 2}{12.7\,hrs} = \lambda_{EC} + \frac{39}{61}\lambda_{EC} = \lambda_{EC}\frac{\times 100}{61}$$

$$\lambda_{EC} = \frac{\cancel{\ln 2} \times 61}{100 \times 12.7\,hrs} = \frac{\cancel{\ln 2}}{T_{\frac{1}{2}EC}}$$

$$T_{\frac{1}{2}EC} = \frac{12.7\,hrs \times 100}{61}$$
$$= 20.81\,hrs$$

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Q. 28: A radioactive substance has a half life of t_0. Two particular nuclei – let's name them as nucleus A and B-have not decayed over a particular observation period of $6t_0$.

(a) What is the probability that A will decay over a further period of $3t_0$?

(b) What is the probability that both A and B will survive over a further period of $3t_0$?

$6t_0$

$N \longrightarrow$

$$N_0 - \frac{N_0}{2} - \frac{N_0}{4} - \frac{N_0}{8}$$

$$\frac{7N_0}{8}$$

$$\frac{\left(\frac{7N_0}{8}\right)}{(N_0)} = \frac{7}{8} \quad (a)$$

(b) $$\frac{(N_0/8)}{N_0} = \frac{1}{8}$$

$$\frac{1}{8}$$

$$\frac{1}{8} \times \frac{1}{8} = \frac{1}{64}$$

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Q. 33: A radioactive nucleus A decay to C after emitting two α and three β particles. Another nucleus B decays to C by emitting one α and five β particles. Half life of A and B are 1hr and 2hr respectively and all intermediates have negligible half lives. At time $t = 0$, population of nuclei of A and B are $4N_0$ and N_0 respectively and population of C is zero.

(a) Find the difference in atomic numbers and mass numbers of A and B.

(b) Find the population of C when both A and B have equal population.

(c) Find time t_0 when both A and B have equal rate of decay.

$${}^{A}_{Z}A \xrightarrow{T_{1/2}=1hr} {}^{A-8}_{Z-1}C + 2\,{}^{4}_{2}He + 3\,{}^{0}_{-1}\beta$$

$${}^{A-4}_{Z-4}B \longrightarrow {}^{A-8}_{Z-1}C + {}_{2}He^{4} + 5\,{}_{-1}\beta^{0}$$

4

$A-(A-4)=4$

$Z-1-3$

$A-8+4+0$

$$\frac{15N_0}{4}+\frac{3N_0}{4} = \frac{18N_0}{4} = \frac{9N_0}{2}$$

$$2^{t} = 4\,2^{t/2}$$

$$2^{t/2} = 4 = 2^{2} \Rightarrow t = 4hr$$

4 half lives

$$A \xrightarrow{T_{1/2}=1hr} C$$

$4N_0$ $\frac{4N_0}{2^{t/1}}$ $\frac{N_0}{4}$

$4N_0 - \frac{N_0}{4}$ $\frac{15N_0}{4}$

$$B \xrightarrow{2hrs} C$$

2 half

N_0 $\frac{N_0}{2^{t/2}}$ $\frac{N_0}{4}$ $\frac{3N_0}{4}$

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$$A_A = A_B$$

$$N_A \lambda_A = N_B \lambda_B$$

$$\frac{N_A}{1\,hr} = \frac{N_B}{2\,hr}$$

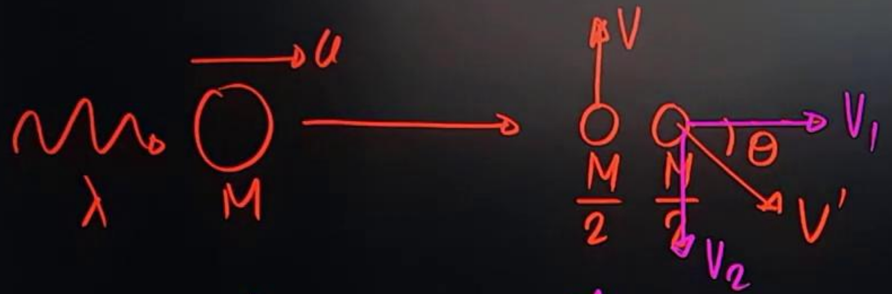
$$\frac{N_B}{N_A} = 2 = \frac{(N_0) \times 2^{t/1}}{2^{t/2} \times 4N_0}$$

$$8 = \frac{2^t}{2^{t/2}} = 2^{t/2} = 2^3$$

$$t/2 = 3$$

$$t = 6\,hr$$

Q. 40: A diatomic molecule moving at a speed u absorbs a photon of wavelength λ and then dissociates into two identical atoms. One of the atoms is found to be moving with a speed v in a direction perpendicular to the initial direction of motion of the molecule. Take mass of the molecule to be M and calculate the binding energy of the molecule. Assume that momentum of absorbed photon is negligible compared to that of the molecule.



Momentum conservation

x dirn: $\cancel{\frac{h}{\lambda}}^{\,0} + Mu = \frac{M}{2}v_1$

$$\boxed{v_1 = 2u}$$

y dirn: $\frac{M}{2}v = \frac{M}{2}\times v_2$

$$\boxed{v_2 = v}$$

$$v' = \sqrt{v^2 + (2u)^2}$$

$$BE = \left(\frac{hc}{\lambda} + \frac{1}{2}Mu^2\right) - \left(\frac{1}{2}\frac{M}{2}v^2 + \frac{1}{2}\frac{M}{2}\times\left(v^2 + (2u)^2\right)\right)$$

$$= \frac{hc}{\lambda} + \frac{1}{2}Mu^2 - \frac{Mv^2}{4} - \frac{Mv^2}{4} - Mu^2$$

$$\boxed{= \frac{hc}{\lambda} - \frac{1}{2}Mu^2 - \frac{1}{2}Mv^2}$$

Q. 45: Two radioactive elements A and B are present in a mixture. Both are beta active. Half lives of A and B are 30 minute and 60 minute respectively. The initial activity of the sample was measured to be 40 decay per second and after an hour only 30 beta particles were detected in an interval of 2 second. Assume that law of radioactivity decay is obeyed.

(a) Find the ratio of number of active nuclei of A to that of B in the mixture initially.

(b) The molar mass of both A and B is closed to 200g. The mixture contains other substance (non radioactive) apart from A and B. What fraction of the entire mixture is radioactive (fraction by weight) if it is given what mass of the mixture is 1μg.

$\boxed{\begin{matrix} A \\ B \end{matrix}}$

$$A = \frac{dN_A}{dt} + \frac{dN_B}{dt}$$

$$= \lambda_A N_A + \lambda_B N_B$$

$t=0 \quad 40 = \lambda_A N_A + \lambda_B N_B$ — (1)

$t = 1\text{hr} \quad \frac{30}{2} = \lambda_A N_A' + \lambda_B N_B' = 15$

$$N_B' = \frac{N_B}{2} \qquad N_A' = \frac{N_A}{4}$$

$$15 = \lambda_A \frac{N_A}{4} + \lambda_B \frac{N_B}{2}$$

$60 = \lambda_A N_A + 2\lambda_B N_B$ — (2)

(2) − (1)

$$\lambda_B N_B = 20 = \lambda_A N_A$$

$$\boxed{\frac{N_A}{N_B} = \frac{\lambda_B}{\lambda_A} = \frac{t_{1/2A}}{t_{1/2B}} = \frac{30}{60} = \frac{1}{2}}$$

$$\lambda = \frac{\ln 2}{t_{1/2}}$$

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$$\lambda_A N_A = 20 = N_B \lambda_B$$

$$N_A = \frac{20}{\lambda_A}$$

$$\boxed{\lambda_A = \frac{\ln 2}{t_{1/2 A}}}$$

$$N_A = \frac{20 \times 30 \times 60}{0.693}$$

$$N_A = \frac{36 \times 10^3}{0.7} = \underline{5.2 \times}$$

$$N_B = 2 \times 5.2 \times 10^4$$

$$N_A + N_B = 10$$

$$n = \frac{15.6 \times 10^4}{6.023 \times 10^{23}}$$

$$m = \frac{15.6 \times 10^4}{6.023 \times 10^{23}} \times 200$$

$$= 2.6 \times 10^{-19} \times 200$$

$$= 5.2 \times 10^{-17}\, g$$

$$= \frac{5.2 \times 10^{-17}}{1 \times 10^{-6}}$$

$$= \underline{5.2 \times 10^{-11}}$$

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Q. 59: Nuclei of radioactive element A are being produced at rate t^2 at time t. The element A has decay constant equal to λ. If N is the number of active nuclei of element A at any time t then $\frac{dN}{dt}$ was found to be minimum at $t = t_0$. Find N at time $t = t_0$.

$$\xrightarrow{t^2} \quad \xrightarrow{\lambda N}$$

$$\Rightarrow f' = 2t - \lambda \frac{dN}{dt}$$

$$0 = 2t_0 - \lambda\left(t_0^2 - \lambda N\right)$$

$$= 2t_0 - \lambda t^2 + \lambda^2 N$$

$$\boxed{N = \frac{\lambda t_0^2 - 2t_0}{\lambda^2}}$$

Q. 60: The power radiated by a star and its mass m are related by $\frac{P}{P_0} = \left(\frac{m}{M_0}\right)^{7/2}$ where P_0 and M_0 are power radiated by the Sun and mass of the Sun respectively. Assume that the fraction of mass lost by the star since its birth is $\alpha (<< 1)$. Calculate the age of the star in terms of α, M_0 P and P_0.

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$$P = P_0\left(\frac{m}{M_0}\right)^{7/2}$$

$$E = \Delta m c^2$$

$$P = \frac{dE}{dt} = -\frac{dm}{dt}c^2$$

$$-\frac{dm}{dt}c^2 = P_0\left(\frac{m}{M_0}\right)^{7/2}$$

$$\frac{dm}{m^{7/2}} = \frac{-P_0\,dt}{M_0^{7/2}c^2}$$

$$\int_{m_0}^{m}\frac{dm}{m^{7/2}} = \int_0^t \frac{-P_0\,dt}{M_0^{7/2}c^2}$$

$$\left[\frac{m^{-7/2+1}}{-5/2}\right]_{m_0}^{m} = \frac{-P_0 t}{M_0^{7/2}c^2}$$

$$\left[m^{-5/2} - m_0^{-5/2}\right] = +\frac{5P_0 t}{2M_0^{7/2}c^2}$$

$$m^{-5/2}\left[1 - \frac{m^{5/2}}{m_0^{5/2}}\right] = (\quad)$$

$$\frac{m_0 - m}{m_0} = \alpha$$

$$1 - \frac{m}{m_0} = \alpha$$

$$\frac{m}{m_0} = 1-\alpha$$

$$\left(\frac{m}{m_0}\right)^{5/2} = (1-\alpha)^{5/2} = 1 - \frac{5}{2}\alpha$$

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$$\frac{1}{m^{5/2}}\left(1-\left(1-\frac{5}{2}\alpha\right)\right) = \cancel{\frac{5}{2}}\frac{P_0 t}{M_0^{7/2}c^2}$$

$$\cancel{\frac{5}{2}}\alpha\cdot\frac{1}{m^{5/2}}$$

$$t = \frac{\alpha M_0^{7/2} c^2}{P_0 m^{5/2}}$$

$$= \frac{\alpha M_0^{\cancel{7/2}} c^2}{P_0 \cancel{M_0^{5/2}}\left(\frac{P}{P_0}\right)^{5/7}} = \frac{\alpha M_0 c^2}{P_0^{2/7} P^{5/7}}$$

$$\frac{P}{P_0} = \left(\frac{m}{M_0}\right)^{7/2}$$

$$M_0\left(\frac{P_0}{P_0}\right)^{2/7} = m$$